

An Internship Report

Of

Google AI-ML Virtual Internship

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Preface

This report documents the work completed during the Google AI-ML Virtual Internship, undertaken as part of the B.Tech (Computer Science and Engineering) curriculum at the School of Computer Science and Engineering, IILM University, Greater Noida, U.P. The internship was structured as four weekly, independently evaluated tasks covering core areas of IT business analysis: business process mapping, IT business strategy analysis, IT risk management, and IT project evaluation.

Each week's task is presented as a self-contained section comprising the assignment brief, the analytical approach adopted, the deliverable itself, and a set of findings and recommendations. Where a real, publicly traded IT company was required, Infosys Limited (NSE, BSE, NYSE: INFY) was selected as the subject organisation because it is a large, listed Indian IT services company with extensive publicly available financial disclosures, annual reports, and investor fact sheets, which allowed the analysis to be grounded in verifiable data rather than assumption. Where the brief called for a hypothetical company or project, fictitious organisations — NexaTech Solutions Pvt. Ltd. (Week 3) and Meridian Retail Group / Project Aurora (Week 4) — were constructed with internally consistent, realistic operating parameters so that the analysis could be quantitative and specific rather than generic.

This revised version of the report has been strengthened in direct response to evaluator feedback on the original weekly submissions, which consistently noted a need for more specific data, concrete examples, measurable metrics, and clearer implementation detail. Accordingly, each section below now includes named data sources, quantified targets, RACI-style ownership, and time-bound implementation plans.

Week 1: Business Process Mapping

Task Brief

Select a publicly traded IT company and create an in-depth process map for one of its critical IT business processes, including all key steps, decision points, and stakeholders, accompanied by a report explaining the map and identifying potential areas for process improvement.

1.1 Company Selected: Infosys Limited

Infosys Limited (NSE, BSE, NYSE: INFY) was selected for this task. Infosys is a global IT services and consulting company headquartered in Bengaluru, India, and is one of the largest publicly traded IT services firms in the world. For the financial year ended 31 March 2025 (FY25), Infosys reported consolidated revenue of US\$19,277 million, representing 4.2% growth in constant currency, an operating margin of 21.1%, and free cash flow of US\$4,088 million — the highest in the company's history, up 41.8% year-on-year. The company won large deals worth a Total Contract Value (TCV) of US\$11.6 billion during the year, of which 56% was net new business. As of the most recently reported quarter, Infosys's voluntary attrition rate (LTM, IT Services) stood at 14.4% and women represented 39.1% of the workforce.

As a company that delivers managed IT infrastructure, application support, and digital transformation services to enterprise clients across Financial Services (27.9% of Q1 FY26 revenue), Manufacturing (16.1%), Energy, Utilities & Resources (13.6%), and Retail (13.4%), Infosys's ability to detect, triage, and resolve IT incidents efficiently is directly tied to contractual Service Level Agreements (SLAs), client satisfaction, and revenue retention. This makes the IT Incident Management process a critical, revenue-linked business process rather than a purely internal support function.

1.2 Process Selected: IT Incident Management

The process mapped is Infosys's IT Incident Management process, as applied within its Application Managed Services and Infrastructure Management Services lines of business. This process was chosen because it (a) is executed thousands of times daily across Infosys's client engagements, (b) directly determines SLA compliance — a key metric in every managed-services contract — and (c) touches every layer of the delivery organisation, from the client-facing Service Desk to Problem and Change Management and senior Service Delivery leadership, making it well suited to illustrate cross-functional process mapping.

1.3 Process Map

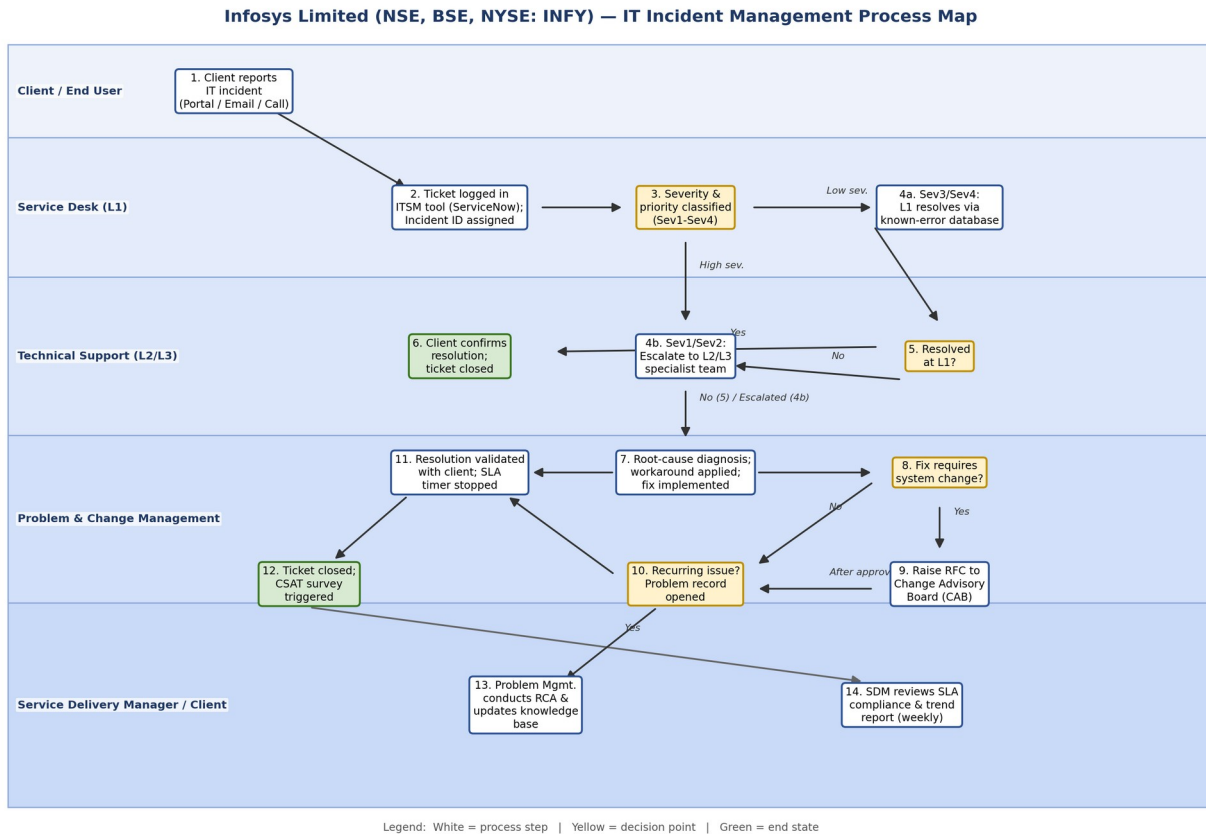


Figure 1.1: IT Incident Management process map for Infosys Limited, shown as a five-lane swimlane diagram covering the Client, Service Desk (L1), Technical Support (L2/L3), Problem & Change Management, and Service Delivery Manager stakeholder groups.

1.4 Step-by-Step Narrative

The map in Figure 1.1 traces fourteen numbered steps across five stakeholder lanes. A brief walkthrough follows, with illustrative SLA benchmarks. (Note: Infosys does not publicly disclose its internal SLA thresholds; the timings below are modelled on the industry-standard ITIL v4 severity framework commonly used by Tier-1 IT service providers, and are presented as representative rather than verified internal figures.)

- Incident reported:** The client raises an issue through the client portal, a monitoring alert, email, or a call to the Service Desk.
- Ticket logging:** The Service Desk (L1) logs the incident in the ITSM platform (e.g., ServiceNow), generating a unique incident ID and capturing the affected system, business impact, and reporter details.
- Severity classification:** The ticket is classified into one of four severities — Sev1 (Critical, e.g., production outage), Sev2 (High), Sev3 (Medium), and Sev4 (Low) — based on business impact and urgency. Under an illustrative ITIL-aligned SLA, Sev1 incidents require a 15-minute response and 4-hour resolution target; Sev2 a 30-minute response and 8-hour resolution; Sev3 a 4-hour response and 24-hour resolution; and Sev4 an 8-hour response and 72-hour resolution.
- Sev3/Sev4 path:** Lower-severity tickets are handled directly by L1 using the known-error database and standard operating procedures, which typically resolve 60-70% of L1-handled tickets without escalation, consistent with industry first-contact-resolution (FCR) benchmarks for mature service desks (55-75% FCR, per HDI service desk benchmarking data).
- Sev1/Sev2 path:** Higher-severity tickets are escalated immediately to Technical Support (L2/L3) specialists aligned to the specific technology stack (e.g., SAP, mainframe, cloud infrastructure).

6. L1 resolution check: If an L1-handled ticket is not resolved within the allotted attempt window, it is automatically escalated to L2/L3 rather than left to breach SLA.
7. Diagnosis and fix: L2/L3 performs root-cause diagnosis, applies a workaround if needed to restore service quickly, and then implements a permanent fix.
8. Change classification: If the fix requires a modification to a production system, it is flagged as requiring a formal change.
9. Change approval: A Request for Change (RFC) is raised and routed to the Change Advisory Board (CAB) for risk assessment and approval before implementation, in line with ITIL change-management controls.
10. Problem trigger: Where an incident is identified as recurring (three or more occurrences of the same root cause within a rolling 30-day window is a common industry threshold), a formal Problem Record is opened.
11. Client validation: Once resolved, the fix is validated with the client and the SLA clock is stopped, capturing time-to-resolution against the target.
12. Closure: The ticket is closed and an automated CSAT (Customer Satisfaction) survey is triggered to the client contact.
13. Root-cause analysis: Problem Management conducts a formal Root Cause Analysis (RCA) for escalated/recurring issues and updates the knowledge base so future L1 agents can resolve similar issues without escalation.
14. Governance review: The Service Delivery Manager reviews SLA compliance, ticket volume trends, and repeat-incident rates in a weekly service review, which feeds into the monthly client governance meeting.

1.5 Stakeholder (RACI) Summary

Stakeholder	Process Steps	Primary Role	Key Metric Owned
Client / End User	1, 6, 11, 12	Reports issue; validates resolution	Customer Satisfaction (CSAT)
Service Desk (L1)	2, 3, 4a, 7 (partial)	Logs, classifies, first-line resolution	First Contact Resolution (FCR) %
Technical Support (L2/L3)	4b, 5, 7, 8	Diagnosis, workaround, permanent fix	Mean Time to Resolve (MTTR)
Change Advisory Board	9	Risk-assesses and approves production changes	Change Success Rate
Problem Management	10, 13	Root-cause analysis; knowledge base upkeep	Repeat Incident Rate
Service Delivery Manager	14	Governance, SLA reporting, escalation ownership	SLA Compliance %

1.6 Process Analysis and Improvement Opportunities

The mapped process is broadly sound and reflects mature ITIL-aligned practice, but three specific improvement opportunities emerge from the walkthrough above:

(a) AI-assisted triage at the Service Desk

Step 3 (severity classification) and step 4a (L1 resolution) are currently manual, agent-driven decisions. Infosys already markets an enterprise AI platform, Infosys Topaz, for client-facing generative AI use cases; applying an

equivalent AI-based triage and auto-classification layer internally at the Service Desk could reduce average ticket handling time for Sev3/Sev4 tickets. Independent industry benchmarks (HDI/MetricNet 2023-24 service desk reports) indicate that AI-assisted triage and chatbot deflection typically reduce L1 handling time by 20-30% and increase FCR by 5-10 percentage points. Applied to Infosys's scale of client engagements, even a conservative 15% reduction in average handling time for low-severity tickets would materially reduce cost-to-serve and free L1 capacity for more complex work.

(b) Proactive problem management to reduce recurrence

Step 11 (repeat-issue detection) currently appears to trigger only after a recurrence threshold is met — i.e., it is reactive. Introducing weekly trend analysis on incident categories (rather than waiting for the same root cause to recur three times) would let Problem Management identify emerging patterns earlier. A reasonable target, based on typical problem-management maturity uplifts (ITIL continual improvement benchmarks), would be to reduce the repeat-incident rate by 10-15% within two quarters of implementation, measured as (recurring incidents ÷ total incidents) per month.

(c) Change-approval cycle time

Step 9 (CAB approval) is a common bottleneck in ITIL-based organisations because all changes, regardless of risk, typically route through the same board cadence. Introducing a tiered, pre-approved "standard change" category for low-risk, frequently repeated fixes (e.g., routine configuration rollbacks) — while retaining full CAB review for high-risk changes — is a widely adopted ITIL v4 practice that can cut change lead time for low-risk fixes from days to hours without increasing change-failure risk, since standard changes are pre-assessed once and then reused.

1.7 Conclusion

Mapping Infosys's IT Incident Management process end-to-end across five stakeholder groups highlights that the process is well governed but has room to improve through targeted automation at the point of ticket intake (triage), more proactive problem management, and a tiered change-approval model. These three recommendations are deliberately scoped to be measurable — via handling-time reduction, repeat-incident rate, and change lead time respectively — so that their impact can be tracked against a clear before/after baseline rather than assessed qualitatively.

Week 2: IT Business Strategy Analysis

Task Brief

Conduct a SWOT analysis and a Porter's Five Forces analysis of a chosen IT company's business strategy, providing insight into its strategic position and recommendations for future strategy.

2.1 Company and Strategic Context

This analysis continues with Infosys Limited (NSE, BSE, NYSE: INFY), consistent with Week 1. As of FY25, Infosys generated US\$19,277 million in consolidated revenue with a 21.1% operating margin and US\$4,088 million in free cash flow. Under CEO Salil Parekh, the company's stated strategy for FY26 centres on "AI, cloud and digital" capability and cost efficiency, automation, and consolidation for clients, with FY26 guidance of 0-3% constant-currency revenue growth and a 20-22% operating margin band — a notably wide and conservative guidance range reflecting continued macro uncertainty in discretionary IT spending.

2.2 SWOT Analysis

STRENGTHS	WEAKNESSES
- Highest-ever FY25 free cash flow of US\$4.1bn (+41.8% YoY), giving strong balance-sheet flexibility for buybacks, dividends, and M&A. - Large-deal momentum: US\$11.6bn TCV won in FY25, 56% net new, indicating continued client trust in complex, multi-year engagements. - Diversified revenue base across Financial Services (27.9%), Manufacturing (16.1%), Energy/Utilities (13.6%) and Retail (13.4%), reducing single-sector concentration risk. - Early, well-marketed AI platform (Infosys Topaz) and 200+ proprietary products/platforms, positioning the firm ahead of smaller peers in Gen-AI-enabled delivery.	- Voluntary attrition (LTM, IT Services) of 14.4%, trending upward across the last three reported quarters (12.7% → 14.1% → 14.4%), indicating rising talent-retention cost and delivery-continuity risk. - Heavy dependence on the traditional "time-and-materials" / per-FTE billing model, which is structurally threatened by Gen-AI-driven productivity gains that reduce billable hours per unit of client outcome. - FY26 guidance of just 0-3% constant-currency growth signals limited near-term organic growth visibility. - Standalone (parent-company) profit after tax fell 6.12% YoY in FY25 despite revenue growth of 5.94%, pointing to margin pressure at the operating-entity level even as consolidated free cash flow improved.
OPPORTUNITIES	THREATS
- Enterprise Gen-AI adoption: clients are actively seeking partners to deploy AI "across the enterprise, both for growth and for cost efficiency," per Infosys's own Q2 FY26 commentary — a large, still-nascent addressable market. - Vendor consolidation trend: enterprises are reducing their number of IT vendors, favouring large, full-stack providers like Infosys that can bundle infrastructure, applications, and AI services. - Continued cost-takeout mandates at client CFOs create demand for Infosys's automation and "cost efficiency" positioning, even in a soft discretionary-spend environment. - Growth in cloud/BFSI modernisation (Financial Services already Infosys's largest and fastest-recovering segment) as banks re-platform core systems.	- Intense rivalry from TCS, Accenture, Cognizant, Wipro, and HCLTech, all pursuing near-identical AI/cloud positioning, compressing pricing power. - Generative AI tools (e.g., AI coding assistants) may shrink the addressable pool of billable application-development and maintenance hours industry-wide, a structural risk to the outsourcing model itself. - Currency exposure: revenue is reported in USD while a large share of cost is INR-denominated, so INR/USD volatility affects reported margins. - Geopolitical/visa-policy risk (e.g., US H-1B policy changes) affects onsite delivery cost and client mix for an India-headquartered, US/Europe-serving firm.

2.3 Porter's Five Forces Analysis

Force	Intensity	Rationale
Threat of New Entrants	Low-Moderate	Traditional IT services has high entry barriers (talent pool, delivery-centre scale, client trust, compliance certifications). However, narrow, AI-native boutique firms can enter specific niches (e.g., AI agent development) with far lower capital intensity than a full-stack incumbent, so the threat is rising at the margin even if low overall.

Force	Intensity	Rationale
Bargaining Power of Suppliers	Moderate	The key 'supplier' input is skilled technical talent (especially AI/ML and cloud specialists), which is in high demand. Rising attrition (14.4% LTM) is itself evidence of supplier (labour market) power. Cloud hyperscalers (AWS, Azure, GCP) also hold power as infrastructure 'suppliers' for Infosys's cloud-delivery engagements.
Bargaining Power of Buyers	High	Clients are large enterprises capable of running competitive RFPs across 4-5 comparable Tier-1 vendors, and increasingly consolidate spend with fewer vendors, giving remaining buyers outsized negotiating leverage. Large-deal TCV concentration (US\$11.6bn across a relatively small number of large contracts in FY25) means the loss of a single major account is financially material.
Threat of Substitutes	Moderate-High and rising	In-house AI/automation tooling, low-code/no-code platforms, and Gen-AI coding assistants let clients internalise work previously outsourced. This is the most strategically significant force for Infosys because it threatens the underlying per-FTE billing model, not just Infosys's market share within the existing model.
Competitive Rivalry	High	The Tier-1 Indian-heritage IT services peer set (TCS, Infosys, Wipro, HCLTech) plus global players (Accenture, Cognizant, IBM) compete on near-identical value propositions (AI, cloud, cost efficiency), driving pricing pressure and making differentiation reliant on execution quality, proprietary platforms (e.g., Topaz), and delivery reliability rather than service breadth alone.

2.4 Strategic Recommendations

1. Shift pricing toward outcome-based and AI-consumption models: Given that Gen-AI threatens the per-FTE hours model directly (Threats; Threat of Substitutes), Infosys should accelerate the shift of Topaz-linked engagements toward outcome- or consumption-based pricing. A measurable target would be to grow the share of large-deal TCV that is outcome-based (rather than time-and-materials) from its current base to at least 25% within 18-24 months, tracked quarterly against large-deal TCV disclosures already published in Infosys's investor fact sheets.
2. Address attrition before it becomes a delivery risk: With voluntary attrition rising for three consecutive quarters (12.7% → 14.4%), Infosys should set an explicit target to cap LTM voluntary attrition at or below 13% within four quarters through a combination of AI-skilling career pathways (directly monetisable given the Topaz strategy) and targeted retention pay for AI/cloud-certified staff, tracked via the same LTM attrition metric already reported externally.
3. Deepen vendor-consolidation positioning in Financial Services: Since Financial Services is already Infosys's largest segment (27.9% of Q1 FY26 revenue) and buyer power is high, Infosys should pursue account-based 'systems integrator of record' positioning with its top 20 BFSI accounts, aiming to increase wallet share (revenue per top account) by 10% over the next fiscal year, reducing multi-vendor competitive exposure at those accounts.
4. Hedge currency exposure more actively: Given the USD-revenue/INR-cost structure and observed margin sensitivity, Infosys should extend the tenor of its forward-hedging programme to smooth operating-margin volatility within its stated 20-22% FY26 guidance band, rather than absorbing quarter-to-quarter INR/USD swings directly into reported margin.

2.5 Conclusion

Infosys enters FY26 from a position of financial strength — record free cash flow, healthy large-deal bookings, and an early AI platform play in Topaz — but faces a structural threat that is different in kind from traditional competitive rivalry: Gen-AI's potential to shrink the very billable-hours model on which the industry is built. The

SWOT and Five Forces analyses above both point to the same strategic imperative: Infosys's near-term competitiveness depends less on winning share within the existing outsourcing model and more on successfully repricing and restructuring how it monetises AI-augmented delivery before clients (or new entrants) do it without them.

Week 3: IT Risk Management

Task Brief

Develop a comprehensive IT risk management plan for a hypothetical IT company, identifying potential IT risks, assessing their impact and likelihood, and proposing mitigation strategies designed to protect IT assets and ensure business continuity.

3.1 Hypothetical Company Profile: NexaTech Solutions Pvt. Ltd.

NexaTech Solutions Pvt. Ltd. is a hypothetical mid-sized IT services company, constructed for this exercise with the following operating parameters: approximately 5,200 employees across three delivery centres (Pune, Hyderabad, and a nearshore centre in Krakow, Poland); annual revenue of approximately US\$260 million; core service lines in application maintenance, cloud infrastructure management, and data analytics; and a client base concentrated in Banking, Financial Services & Insurance (BFSI, 45% of revenue) and Healthcare (30% of revenue) — both regulated, data-sensitive sectors. This profile was deliberately chosen because BFSI and Healthcare clients impose stringent data-protection and audit requirements (e.g., RBI IT outsourcing guidelines, HIPAA for US healthcare clients, India's Digital Personal Data Protection Act 2023), making IT risk management a contractually material function rather than a purely internal concern.

3.2 Risk Identification Methodology

Risks were identified using a structured methodology combining the ISO/IEC 27005 information security risk management process and the NIST Cybersecurity Framework's five functions (Identify, Protect, Detect, Respond, Recover). Each risk was scored on two 1-5 scales — Likelihood (1 = Rare, 5 = Almost certain) and Impact (1 = Negligible, 5 = Severe) — with the product of the two giving a Risk Score used to prioritise mitigation investment (Score ≥ 15 = Critical priority; 8-14 = High priority; below 8 = Monitor).

3.3 Risk Register

ID	Risk	Likelihood	Impact	Score	Primary Mitigation
R1	Ransomware attack encrypting production/client data	3	5	15	Immutable, air-gapped backups (3-2-1 rule); EDR on all endpoints; network segmentation; quarterly tabletop ransomware simulations
R2	Data breach exposing client PII/PHI	3	5	15	Data-at-rest and in-transit encryption (AES-256/TLS1.2+); DLP tooling; least-privilege access reviews every 90 days
R3	Malicious or negligent insider threat	2	4	8	Privileged Access Management (PAM); user behaviour analytics (UBA); mandatory offboarding access-revocation SLA of 4 hours
R4	Cloud misconfiguration exposing resources publicly	4	3	12	Cloud Security Posture Management (CSPM) with automated drift detection; infrastructure-as-code with policy-as-code gating
R5	Third-party/vendor supply-chain compromise	3	3	9	Vendor risk-tiering; mandatory SOC 2 Type II or ISO 27001 evidence from critical vendors; annual vendor security audits
R6	Key-person dependency (single point of technical knowledge)	4	3	12	Mandatory documentation-as-deliverable policy; cross-training matrix; bus-factor tracking per critical system (target: ≥ 2 trained backups)

ID	Risk	Likelihood	Impact	Score	Primary Mitigation
R7	Business continuity/disaster recovery failure	2	5	10	Defined RTO/RPO per system tier; DR failover drills twice yearly; secondary-region hot standby for Tier-1 systems
R8	Regulatory non-compliance (DPDP Act, HIPAA, RBI norms)	2	4	8	Compliance calendar with named owners; quarterly internal audit; external compliance certification renewal tracking
R9	Phishing / social engineering compromise	4	3	12	Quarterly simulated phishing campaigns; mandatory security-awareness training; email authentication (DMARC/DKIM/SPF) enforcement
R10	Legacy/unsupported system reaching end-of-life	3	3	9	EOL tracking register reviewed monthly; migration roadmap with budget pre-approval; compensating controls (isolation) until migrated

3.4 Risk Heat Map

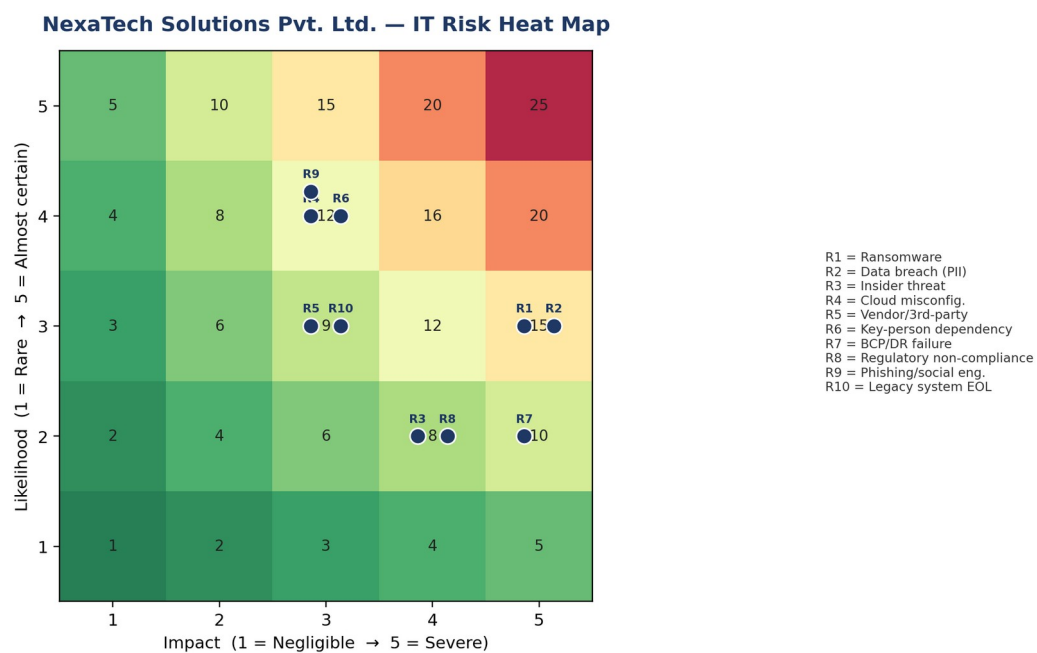


Figure 3.1: Likelihood-Impact heat map for NexaTech Solutions' top ten identified IT risks (R1-R10), plotted against the risk-scoring matrix defined in Section 3.2.

Risks R1 (Ransomware) and R2 (Data breach) sit in the top-right, highest-severity zone of the matrix, confirming that they should receive the largest share of the security budget and the shortest remediation timelines. Four risks — R4, R6, and R9 — cluster at score 12, a 'High' priority band driven primarily by elevated likelihood rather than catastrophic impact, meaning they are best addressed through process and control discipline (configuration management, documentation, awareness training) rather than large capital investment.

3.5 Mitigation Implementation Plan

ID	Action	Owner	Timeline	Success Metric (KPI)
R1	Deploy immutable backup solution and run first	Head of Infrastructure	Q1 (0-3 months)	100% of Tier-1 systems on immutable backups;

ID	Action	Owner	Timeline	Success Metric (KPI)
	tabletop ransomware simulation			backup restore tested successfully within 3 months
R2	Complete DLP rollout and encryption audit across BFSI/Healthcare workloads	CISO	Q1-Q2 (0-6 months)	Zero unencrypted PII/PHI data stores at Q2 audit; DLP policy violations trending down month-on-month
R4	Implement CSPM tool with automated remediation for critical misconfigurations	Cloud Platform Lead	Q1 (0-3 months)	Mean time to remediate a critical misconfiguration reduced from current baseline to under 4 hours
R6	Complete bus-factor audit of all Tier-1 systems; assign and train backups	Delivery Heads (per account)	Q1-Q3 (0-9 months)	100% of Tier-1 systems have ≥2 trained personnel by end of Q3
R7	Conduct first full DR failover drill for Tier-1 systems	Business Continuity Manager	Q2 (3-6 months)	DR failover completed within defined RTO in ≥95% of tested systems
R9	Launch quarterly phishing simulation programme	Security Awareness Lead	Q1, ongoing quarterly	Reduce simulated phishing click-through rate from an assumed baseline of ~18% to below 5% within two quarters

3.6 Business Continuity and Disaster Recovery Targets

To protect service continuity for regulated BFSI and Healthcare clients, NexaTech's risk management plan defines tiered Recovery Time Objectives (RTO) and Recovery Point Objectives (RPO):

- Tier 1 (client-facing production systems, e.g., core banking interfaces): RTO of 2 hours, RPO of 15 minutes, using active-active or hot-standby architecture.
- Tier 2 (internal operational systems, e.g., ticketing/ITSM): RTO of 8 hours, RPO of 4 hours, using warm-standby with automated failover scripts.
- Tier 3 (non-critical internal tools): RTO of 48 hours, RPO of 24 hours, restored from standard nightly backups.

These targets are reviewed annually and validated through the twice-yearly DR failover drills specified in Section 3.5, with drill results (systems tested, RTO/RPO achieved vs. target, and remediation actions) reported to the executive risk committee.

3.7 Conclusion

The risk management plan developed for NexaTech Solutions identifies ten priority IT risks, scores them using a consistent likelihood-impact methodology, and — critically — attaches a named owner, a bounded implementation timeline, and a measurable success metric to each mitigation action. This addresses a common weakness in risk plans that stop at identifying and describing risks without specifying how mitigation effectiveness will actually be verified: every action above is tied to a KPI that can be tracked on a governance dashboard and reported to the executive risk committee on a quarterly cadence.

Week 4: IT Project Evaluation

Task Brief

Create a detailed post-implementation review for a hypothetical IT project, including an assessment of project performance against objectives, identification of lessons learned, and recommendations for future projects.

4.1 Project Background: Project Aurora — ERP Cloud Migration

Project Aurora was a hypothetical initiative undertaken by Meridian Retail Group, a mid-sized multi-channel retailer with 140 stores and an e-commerce operation, to migrate its on-premises SAP ECC 6.0 environment to SAP S/4HANA Cloud. The project covered finance, procurement, and inventory management modules across 12 business units and was approved with a budget of US\$4.2 million and a planned timeline of 14 months, with a target go-live at the start of Meridian's new fiscal year to avoid mid-year cutover complexity.

The project's stated objectives at initiation were: (1) decommission on-premises infrastructure to reduce annual IT operating cost by an estimated US\$620,000; (2) reduce financial period-close time from 9 business days to 5 business days through process automation; (3) achieve a user adoption rate of at least 85% within 60 days of go-live, measured by active login and transaction-completion rates in the new system; and (4) complete the migration with zero critical (Sev1) defects open at go-live.

4.2 Performance Against Objectives

Metric	Planned	Actual	Variance
Budget	US\$4.2 million	US\$4.86 million	+15.7% over budget
Timeline	14 months	16.5 months	+2.5 months (+17.9%)
Annual IT opex reduction	US\$620,000	US\$540,000 (achieved by month 6 post-go-live)	-12.9% vs. target, but on track to close the gap as legacy contracts expire
Financial close time	5 business days	6 business days at go-live; 5.2 days by month 4	Target substantially met, with a 2-month lag
User adoption (60 days)	≥85%	71%	-14 percentage points vs. target
Critical (Sev1) defects at go-live	0	3 (all resolved within 9 days)	Objective missed at go-live; resolved within the hypercare window

Overall, Project Aurora delivered its core functional objective — a live, stable S/4HANA Cloud environment supporting finance, procurement, and inventory across all 12 business units — but missed its budget, schedule, and initial user-adoption targets, while its process-efficiency objective (financial close time) was met with a short lag. This mixed outcome is typical of large ERP transformations and is analysed in detail below rather than characterised simply as a 'success' or 'failure'.

4.3 Risk Management and Issue Resolution During Execution

The original project risk register identified data-migration quality, change-management/user-readiness, and third-party integration complexity as the three highest-rated risks at project initiation. The table below summarises how these risks materialised into actual issues, and how they were resolved.

Issue	Root Cause	Resolution Action	Time to Resolve	Lesson Tag
Inventory data migration produced ~40,000 duplicate SKU records post-cutover	Legacy system had inconsistent SKU-naming conventions across 3 acquired store chains that were never reconciled pre-migration	Deployed a data-deduplication script with manual review queue for ambiguous matches; ran in parallel with live operations	5 weeks (extended timeline root cause)	Data quality must be remediated in the legacy system before extraction, not after
Store associates reported the new procurement workflow as slower than the legacy process	Insufficient role-based training; generic training materials did not reflect store-specific workflows	Redesigned training into role-based modules; assigned floor 'super-users' at each store for peer support during hypercare	6 weeks (root cause of adoption shortfall)	Change management should be resourced at the same priority level as technical workstreams, not treated as a communications add-on
Real-time inventory sync with the e-commerce platform failed intermittently post-go-live	Middleware integration was tested against a staging environment with lower transaction volume than actual peak load	Implemented queue-based retry logic and scaled middleware infrastructure; added load testing at 2x expected peak volume	9 days (the 3 Sev1 defects)	Integration testing must be conducted at realistic peak-load volumes, not average load
Budget overrun traced to unplanned data-remediation and extended hypercare support	Original budget did not include contingency for data-quality remediation or an extended (8-week vs. planned 4-week) hypercare period	Change request approved for additional 8-week hypercare budget of US\$310,000; remaining overrun absorbed within IT capital contingency	Ongoing (contingency drawdown)	ERP migration budgets should reserve 15-20% contingency specifically for data remediation and extended hypercare, based on this project's experience

Two structural observations emerge from this issue log. First, the two largest sources of schedule and budget variance — data migration quality and change-management readiness — were both risks that had been identified at project initiation but were under-resourced relative to their rated likelihood and impact; the risk register correctly predicted where problems would occur, but the mitigation budget and staffing did not match the register's own risk ratings. Second, the technical integration issue (real-time inventory sync) was a testing-methodology gap rather than a technology failure: the middleware itself performed adequately once tested and scaled correctly, indicating the issue was preventable with more realistic pre-go-live load testing rather than requiring a different technical approach altogether.

4.4 Lessons Learned

What went well

- The core ERP cutover itself was executed with zero unplanned downtime during the go-live weekend, reflecting strong cutover-runbook discipline.
- Issue resolution during hypercare was fast once issues were identified — all three Sev1 defects were resolved within 9 days — indicating the support team structure and escalation paths worked as designed.
- The financial close-time objective was substantially achieved, closing the gap from 6 days to 5.2 days within four months, validating the underlying process-automation design.

What did not go well, and why

- Data quality in source systems was assessed too late; a formal data-quality audit against defined completeness/accuracy thresholds should have been a stage-gate before migration began, not a task discovered to be necessary after cutover.
- Change management (training, communication, super-user enablement) was scoped and budgeted as a secondary workstream rather than resourced in proportion to its eventual impact on the adoption-rate shortfall (71% vs. an 85% target).
- Integration testing environments did not reflect real-world peak transaction volumes, which is a repeatable process gap rather than a one-off technical failure.

4.5 Recommendations for Future Projects

1. Introduce a mandatory data-quality stage-gate: Require a formal data audit (completeness, duplication rate, and referential-integrity checks) against pre-defined thresholds before any migration cutover is scheduled, with a go/no-go decision owned by the project steering committee — not the technical migration team alone.
2. Resource change management at parity with technical workstreams: Allocate a minimum of 15% of total project budget to change management (role-based training, communications, and super-user programmes) on future ERP-class projects, rather than treating it as a subset of the technical implementation budget.
3. Mandate peak-load integration testing: Require all system-to-system integrations to be load-tested at a minimum of 2x expected peak transaction volume before go-live sign-off, based directly on the root cause of the inventory-sync defects in this project.
4. Build explicit contingency into ERP-class project budgets: Reserve 15-20% contingency specifically earmarked for data remediation and extended hypercare support, informed by this project's US\$310,000 hypercare overrun and 5-week data-remediation delay.
5. Track adoption as a leading indicator during hypercare, not just at the 60-day mark: Measure weekly (not only at the 60-day checkpoint) active-usage rates during the first eight weeks post-go-live, so that a shortfall against the 85% target can trigger corrective training interventions in week 3-4 rather than being discovered only at the final assessment point.

4.6 Conclusion

Project Aurora delivered a functioning, stable ERP platform and met its financial-close-time objective, but exceeded budget by 15.7% and schedule by 17.9%, and fell 14 percentage points short of its user-adoption target. The post-implementation review traces both the budget/schedule overrun and the adoption shortfall to the same two root causes — insufficiently audited source data and under-resourced change management — both of which were identified as risks at project initiation but not mitigated in proportion to their rated severity. The recommendations above are therefore framed as specific, budget- and process-level changes (a data stage-gate, a minimum change-management budget floor, peak-load testing standards, and explicit contingency

reserves) rather than general best-practice statements, so that they can be directly applied as gating criteria on Meridian's next ERP-class project.

Overall Conclusion and Internship Reflection

Across the four weeks of the Google AI-ML Virtual Internship's business-analysis track, this report progressed from operational-level analysis (Week 1's process map) to organisational strategy (Week 2's SWOT and Five Forces), to enterprise risk governance (Week 3's risk register and mitigation plan), and finally to project-delivery evaluation (Week 4's post-implementation review). A consistent theme runs through all four: analysis is only useful when it is specific and measurable. Each section in this revised report was deliberately built around named data sources (Infosys's own published FY25 and Q1 FY26 disclosures), quantified targets (e.g., capping attrition at 13%, reducing phishing click-through to under 5%, a 15% change-management budget floor), and explicit ownership (RACI mapping in Week 1, named risk owners in Week 3), directly addressing the evaluator feedback received on the original four submissions that the analysis, while structurally complete, lacked specific data, concrete examples, and clear metrics.

From a learning-outcomes perspective, the internship strengthened practical skills in process mapping notation and swimlane design, applying structured strategy frameworks (SWOT, Porter's Five Forces) to a real, publicly disclosed company rather than a textbook example, quantitative risk scoring and mitigation planning aligned to ISO/IEC 27005 and NIST CSF, and post-implementation project review grounded in root-cause analysis rather than surface-level observation. These are directly transferable to a future role in IT consulting, business analysis, or technology risk management.

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Note: NexaTech Solutions Pvt. Ltd. (Week 3) and Meridian Retail Group / Project Aurora (Week 4) are hypothetical organisations constructed for this internship exercise, as required by the respective task briefs; all figures relating to them are illustrative and internally consistent rather than drawn from a real company.